

Streamflow reconstruction in Southern Italy: Hydrological Models versus Machine Learning Approaches

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Abstract

The availability of continuous long-term hydrological records is essential for improving the understanding of hydro-climatic processes and supporting sustainable water resources management. However, monitoring networks often suffer from spatial and temporal gaps that limit dataset completeness and reliability. This issue is particularly critical for streamflow data, which is not directly measured but generally derived through rating curves requiring periodic updating over time.

This study focuses on four river basins in the Campania region (southern Italy), located within the Mediterranean climate domain and characterized by marked seasonal variability. In the early 2000s, the hydroclimatic monitoring network of the Campania region underwent a substantial restructuring process, since then observed streamflow data have no longer been available. The analysis considered four major regional catchments: Tanagro, Sele, Calore Irpino, and Volturno, selected based on the historical consistency of their monitoring stations despite the network restructuring. The dataset consists of daily observations of precipitation, temperature and discharge.

This study investigates the potential of machine learning approaches as an alternative to locally calibrated process-based hydrological models for streamflow reconstruction in data-scarce contexts. As a preliminary step, an exploratory analysis based on Self-Organizing Maps (SOM) and K-means clustering was performed to characterise inter-catchment hydroclimatic relationships and identify latent patterns within available observations, providing a basis for the development of a transferable regionalization scheme. Two rainfall-runoff models, TOPMODEL and IHACRES, were implemented and locally calibrated for each catchment, while two machine learning algorithms, Random Forest and XGBoost, were trained within a leave-one-out regionalization framework, in which information from donor basins is used to reconstruct streamflow at the target catchment. The comparative evaluation of these approaches aims to provide insights into the transferability of data-driven methods and their suitability for operational streamflow reconstruction.